

Important Developers of Mechanism Science

This is a direct adaptation of Chapter 3, Important Developers of Mechanism Science and Engineering, “A Bulleted/Pictorial History of Mechanisms and Machines” by Dr. Robert L. Williams II, Professor Mechanical Engineering, Ohio University.

For more extensive lists, visit

- <https://www.iri.upc.edu/people/thomas/repository.html> (A Century of Research on Kinematics (1870-1970): A Bibliographic Compilation) and
- <https://www.dmg-lib.org/dmglib/main/portal.jsp> (Digital Mechanism and Gear Library)



Archimedes (287 – 212 BCE)

- Studied the lever using geometry.
- Until the 1500s CE, Archimedes and Hero of Alexandria (circa 10 – 70 CE) were still the sources of mechanism theory.

Leonardo da Vinci (1452 – 1519)

- Rejuvenated interest and practical design methods in machines and mechanisms during the European Renaissance.
- Some of the more famous Leonardo manuscripts that contain machines are the *Codex Atlanticus* in Milan, *Manuscript B* in Paris, and the *Codex Madrid I* in Madrid, Spain.



James Watt (1736 – 1819)

- In 1782, invented the six-bar Watt's linkage to convert rotational shaft motion into translating motion.
- His linkage generates only approximate straight-line motion.

James J. Sylvester (1814 – 1897)

- Expounded on the Peaucellier Linkage (invented in 1864), to generate an exact straight line from a rotating input.
- Also famous for matrix theory developments.



Alfred B. Kempe (1849 – 1922)

- Extended linkages to trace a given algebraic curve.
- His work led to research between geometry and computer science.

Franz Grashof (1826 – 1893)

- He was a German professor of applied mechanics at the Technische Hochschule Karlsruhe.
- Developed his famous law to determine the rotatability of the input / output links of the four-bar linkage.
- Also responsible for the Grashof Number in free-convection heat transfer.





Franz Reuleaux (1829 – 1905)

Alexander B. W. Kennedy (1847 – 1928)



Ludwig E. H. Burmester (1840 – 1927)



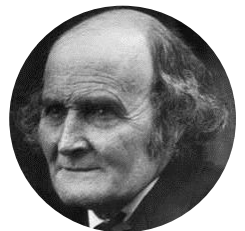
Pafnuty Lvovich Chebyshev (1821 – 1894)



Samuel Roberts (1827 – 1913)



Arthur Cayley (1821 – 1895)



Ferdinand Freudenstein (1926 – 2006)



George N. Sandor (1929 – 1996)



Jaques Denavit (1930 – 2011)



Richard S. Hartenberg (1907 – 1997)



Prof. Arthur G. Erdman



Prof. Francis C. Moon



Prof. John Michael McCarthy

